TIRTHARAJ BHATTACHARYA

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LinkedIn | GitHub | Portfolio | Detailed Profile

PROFESSIONAL SUMMARY

B.Tech (ECE) undergraduate with interests in Embedded Systems, AI/ML, IoT, and intelligent software systems. Hands-on experience developing embedded, IoT, computer vision, and AI-powered applications. Passionate about building intelligent solutions at the intersection of hardware and software.

EDUCATION

- **Guru Nanak Institute of Technology**
  B.Tech in Electronics & Communication Engineering | 2024–Present
- **Central Model School, Barrackpore**
  Class XII (PCMB) – 90%
  Class X – 94%

EXPERIENCE

- **AI/ML Trainee | Ardent Computech Pvt Ltd**
  March 2026 – April 2026
  - Built computer vision and machine learning projects using Python and OpenCV.
  - Performed data analysis and implemented machine learning workflows using TensorFlow.
- **Content Writer | Euphonous X Zyzzva**
  Volunteer | 2025–2026
  - Authored social media content and promotional posts for the institute's literary festival.
  - Prepared event reports and editorial content as part of the Media & Content Team.

SKILLS

Programming: Python, C language

AI/ML: TensorFlow, OpenCV, NumPy

Embedded & IoT: ESP32, Arduino, IoT Prototyping

Core Competencies: Public Speaking, Technical Communication, Leadership, Team Collaboration, Content Writing

PROJECTS

- **NeuroKavach | Ideathon Project (2nd Position)**
  - Conceptualized an AI-enabled wearable system for real-time OCD intervention.
  - Designed the solution concept and contributed to technical presentation and pitch development.
- **Digital Fort Knox | ESP32-Based Security System**
  - Built a device monitoring solution to identify trusted devices and flag unauthorized network access.
  - Designed and prototyped key features for an IoT-based embedded security system.
- **IoT Fire Alert & Automatic Sprinkler System | ESP32-Based Fire Alert and Notification System**
  - Integrated an IR flame sensor, relay-controlled water pump, LED, and buzzer for automated fire response.
  - Enabled Wi-Fi connectivity with **Blynk IoT** for real-time smartphone notifications.
- **AI Study Assistant**
  - Built an AI study assistant using Python, Streamlit, and Gemini API for personalized learning.
  - Implemented AI chat, quiz generation, study planning, and performance analysis.

ACHIEVEMENTS

Leadership: Vice President — GNIT Literary Club (2025-Present)

Research: Co-author of “Design And Development Of A Resonance-Tuned Hybrid Piezoelectric-Electromagnetic Energy Harvesting System” published in IJCRT.

Competitions: • 2nd Position — Idea Manthan (NeuroKavach) • Selected at institute-level screening for Smart India Hackathon